

BANARAS HINDU UNIVERSITY

B.A. / B.Sc. (Hons.) — Semester IV (NEP) Examination, 2025-26

MATHEMATICS**Paper Code: MAT-MJ-43 / MAT-MN-41 : Differential Equations**

Time: 3 Hours

Maximum Marks: 70

REF NO. Conf/Even/N/2025-26/066

*Instructions: Answer **five** questions including Question No. 1 which is compulsory. All questions carry equal marks (14 marks each). Prime (') denotes derivative with respect to x . ODE and PDE denote Ordinary and Partial Differential Equations respectively.*

Question 1. (Compulsory, 14 Marks Total)

- (a) Define the order and degree of an ODE. Determine the order and degree of

$$\left(1 + \left(\frac{dy}{dx}\right)^2\right)^{3/2} = \frac{d^2y}{dx^2}.$$

[2]

- (b) Find
- a
- and
- b
- in the ODE:
- $y'' + ay' + by = 0$
- , if its two linearly independent solutions are
- e^{2x}
- and
- xe^{2x}
- . [2]

- (c) Consider the second order ODE:
- $3xy'' + 2y' + x^2y = 0$
- for which a series solution is being sought by Frobenius method, about
- $x = 0$
- . If the corresponding indicial equation is
- $f(r) = 0$
- , then find
- $f(r)$
- . [2]

- (d) Distinguish between the general solution and the particular solution of an ODE. [2]

- (e) Define linear, semilinear, quasilinear and nonlinear first order PDE's with one example of each type. [2]

- (f) Form the PDE by eliminating the arbitrary function from
- $z = x^n \phi(y/x)$
- . [2]

- (g) Verify that any function of the form
- $\psi(x - ct)$
- satisfies the equation

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2}.$$

[2]

Question 2.

- (a) In a university hostel, suppose the number of occupied rooms
- $N(t)$
- decreases at a rate proportional to the number of unoccupied rooms. If the hostel has 500 rooms and initially 300 are occupied, form the differential equation and find
- $N(t)$
- . Furthermore, if the number of occupied rooms in the second year is 400, then find the number of occupied rooms after 8 years. [6]

- (b) Examine whether the differential equation

$$(f(y))^2 \frac{dx}{dy} + 3f(y)f'(y)x = f'(y)$$

is exact, where f' denotes derivative of f with respect to y . If it is not exact, derive a suitable integrating factor depending only on y to make it exact (Do not use formula). Hence solve the differential equation completely. [8]

Question 3.

- (a) Find the general solution of the ODE:

$$xy'' - (2x + 1)y' + (x + 1)y = \frac{2e^x}{x^2}.$$

Given that $y = e^x$ is one solution of the corresponding homogeneous equation. [6]

- (b) (i) Find the general solution of

$$(x + 2)^2 y'' - (x + 2)y' - 3y = 2 \ln(x + 2), \quad x > -2.$$

[4]

- (ii) Find the solution of the initial value problem $4y'' - y = 0$; $y(0) = 2$, $y'(0) = \beta$. Then find β such that $y(x) \rightarrow 0$ as $x \rightarrow \infty$. [4]

Question 4.

- (a) Consider the differential equation $(1 - x^2)y'' - 2xy' + 20y = 0$. Show that $x = 0$ is an ordinary point of this differential equation and find two linearly independent power series solutions in powers of x . Furthermore, show that one of the solutions is a polynomial of degree 4. [6]
- (b) Define regular singular point of a second order ODE. Use the method of Frobenius to find solutions of the differential equation $2x^2y'' + xy' + (x^2 - 3)y = 0$ in some interval $0 < x < R$. [8]

Question 5.

- (a) Find the equation of the integral surface of the PDE:

$$x^3p + y(3x^2 + y)q = z(2x^2 + y)$$

which passes through the curve $x = 1, z = y^2 + y$, where $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$. [6]

- (b) (i) Show that the equations $f(x, y, p, q) = 0, g(x, y, p, q) = 0$ are compatible if

$$\frac{\partial(f, g)}{\partial(x, p)} + \frac{\partial(f, g)}{\partial(y, q)} = 0.$$

Hence, find the solution of the equations

$$xp - yq = x, \quad x^2p + qz = x + x^2$$

$$\text{where } p = \frac{\partial z}{\partial x} \text{ and } q = \frac{\partial z}{\partial y}. \quad [4]$$

$$(ii) \text{ Use Charpit's method to find the complete integral of } x^2p^2 + y^2q^2 - 4 = 0. \quad [4]$$

Question 6.

(a) Solve the differential equation $(D^2 + DD' - 2D'^2 - 3D + 3D')z = x$, where $D = \frac{\partial}{\partial x}$, $D' = \frac{\partial}{\partial y}$. Find two distinct particular integrals and show that their difference satisfies the corresponding homogeneous equation. [6]

(b) Derive a method to find particular integral of homogeneous linear PDE: $F(D, D')z = f(x, y)$. Use this method to solve

$$(D^2 - 4DD' + 4D'^2)z = e^{2x+y}. \quad [8]$$

Question 7.

(a) Solve the differential equation $(x^2D^2 + 2xyDD' + y^2D'^2)z = x^2y^2$, where $D = \frac{\partial}{\partial x}$, $D' = \frac{\partial}{\partial y}$. [6]

(b) Define singular solution of a first order PDE. Find the singular solution of

$$z = px + qy + \sqrt{1 + p^2 + q^2}. \quad [8]$$
